MIXED REALITY FPV DRONE

This project bridges the gap between legacy analog drone technology and modern Mixed Reality (MR) headsets. By repurposing a consumer Meta Quest 2, we aim to democratize access to First-Person View (FPV) flying, offering a cost-effective alternative to expensive dedicated goggles. Our solution directly interfaces a standard 5.8GHz analog receiver with the headset’s Android operating system, providing pilots with an immersive, large-screen display within a mixed reality environment. Furthermore, by extracting both the live drone feed and the headset's "Passthrough" view, it enables simultaneous external supervision and seamless recording of the pilot's physical and virtual surroundings.

5.8GHz analog communication

Merge in Mixed Reality Application

FPV Drone Stream

Backgroung from headset passthrough view

Contribution

- Hardware Accessibility: Validation of a functional transmission chain that transforms a Quest 2 into FPV goggles for under €50 (receiver cost), compared to €200+ for dedicated analog equipment.

- Hybrid User Experience: Integration of the drone feed into a floating MR window, enabling hand-tracking to manage the interface. Furthermore, the Passthrough feature maintains spatial awareness, conceptually removing the need for a dedicated visual spotter (mandatory under French FPV regulations).

Technical Details

System Architecture:

- Transmitter: BetaFPV Meteor 75 Pro Tinywhoop Drone (Analog Camera + 5.8GHz VTX).
- Receiver: Eachine ROTG02 (Skew Planar Antennas + 5.8GHz RX -> Analog/Digital Converter).
- Processing: Meta Quest 2 (Android OS). The ROTG02 is mounted as /dev/video0 (External Webcam). The headset captures the physical surrounding and merges both videos.
- Radio Controller: BetaFPV Literadio 4

Future development will focus on hardware consolidation by fully integrating the analog receiver directly into the headset. To further enhance immersion and eliminate the need for a physical radio controller, we plan to implement a virtual flight interface driven entirely by native hand-tracking capabilities. Ultimately, this will reduce the entire FPV ground station to a single, standalone wearable device.

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